Duncan Park

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EDUCATION

Cornell University, College of Arts and Sciences — **Ithaca, NY**
B.A. in Mathematics and Computer Science — *Expected May 2029*
Courses: Data Structures and Functional Programming, Real Analysis, Mathematical Foundations of CS — GPA: 3.67/4.0
Stuyvesant High School • *Diploma received June 2025* — **New York, NY**

TECHNICAL SKILLS

Languages: Python, SQL, Java, OCaml, R
ML & Data Science: PyTorch, scikit-learn, pandas, NumPy, SciPy, tidyverse, ggplot2, Autoencoders, Representation Learning
Bioinformatics & Stats: Mixed-Effect Models, Survival Analysis, ANOVA, MAFFT, IQ-TREE, BEAST2
Dev Tools: Git, GitHub, VS Code, IntelliJ, RStudio, Jupyter Notebook

RELEVANT EXPERIENCE

MA'AYAN LAB - COMPUTATIONAL SYSTEMS BIOLOGY AT MOUNT SINAI — **New York, NY**
Summer Trainee — *6/2026 – 8/2026*

- Developed a PyTorch multimodal encoder integrating chemical fingerprints and LINCS gene-expression signatures to predict 83 adverse drug reactions (ADRs); outperformed a chemical-only baseline.
- Engineered preprocessing and representation pipelines for 16M+ FAERS patient reports; standardized 20K+ drug names via PubChem and encoded multi-drug combinations of up to five drugs.
- Applied representation learning and neural networks using Python, PyTorch, NumPy, and SciPy; presented methodology, model results, and biological findings to Mount Sinai faculty at the 2026 Biomedical Big Data Science Symposium.

BIG RED SPORTS NETWORK - FOOTBALL ANALYTICS — **Ithaca, NY**
Analyst, Research and Development Sub-Team Member — *8/2025 – Present*

- Produced 10 weekly scouting reports for Cornell football by analyzing tendencies and visualizing data in heat maps/tables.
- Collaborated with R&D team to build tempo and punting performance models from scratch to track performance metrics.

CORNELL SPORTS ANALYTICS CLUB — **Ithaca, NY**
Social Chair, Web Developer, Analyst — *8/2025 – Present*

- Built and refined ML model predicting field-goal outcomes in the NFL and presented findings to 30 members and leaders.
- Launched club's public website (Jekyll, GitHub Pages, custom domain) with organized archive of 20+ past projects.

PROJECT EXPERIENCE

ANALYZING IMPACT OF WORKLOAD ON INJURY RISK IN THE NFL — **Ithaca, NY**
STSCI 4981 - Sports Analytics in Practice, Cornell University — *3/2026 – 5/2026*

- Used nflreadpy to extract NFL player and schedule data to engineer workload and efficiency features for skill positions.
- Built linear mixed-effect models and survival curves in R to assess how efficiency metrics evolved across workload groups.
- Conducted ANOVA and Chi-squared tests in R that showed high touch volume does not signify short-term injury risk.

MARKOV RANKINGS ACROSS SPORTS LEAGUES — **New York, NY**
Linear Algebra, Stuyvesant High School — *4/2025 – 6/2025*

- Built stochastic matrices using PageRank concepts in R to model steady-state team ratings in the NFL, NHL, and NBA.
- Engineered Monte Carlo simulation engine with Softmax probabilities and home-field adjustments to forecast outcomes.
- Evaluated predictive accuracy against actual results, achieving Normalized Mean Rank Error of 0.0968 for NHL and NFL.

CORNELL DATA SCIENCE PROJECT TEAM — **Ithaca, NY**
Machine Learning Engineer — *9/2026 – Present*

- Selected as 1 of 3 Machine Learning Engineers for Fall 2026 cohort; participating in machine/deep learning training.

LEADERSHIP

NYC TEENS FOR 9 MILLION REASONS • *FOOD PANTRY* — **Queens, NY**
Co-*President, Volunteer* — *9/2022 – 6/2025*

- Founded and led citywide 40+ teen initiative distributing food and clothes at NYC's largest food pantry every Saturday.

AWARDS AND INTERESTS

Awards: Wingate Memorial Award (2025), National Latin Exam Silver Medal (2024, 2025), AP Scholar with Honor (2024)
Interests: *New York Times* Games, Sports (Football, Soccer, Hockey, Skiing), Korean Culture, Cooking, Mahjong